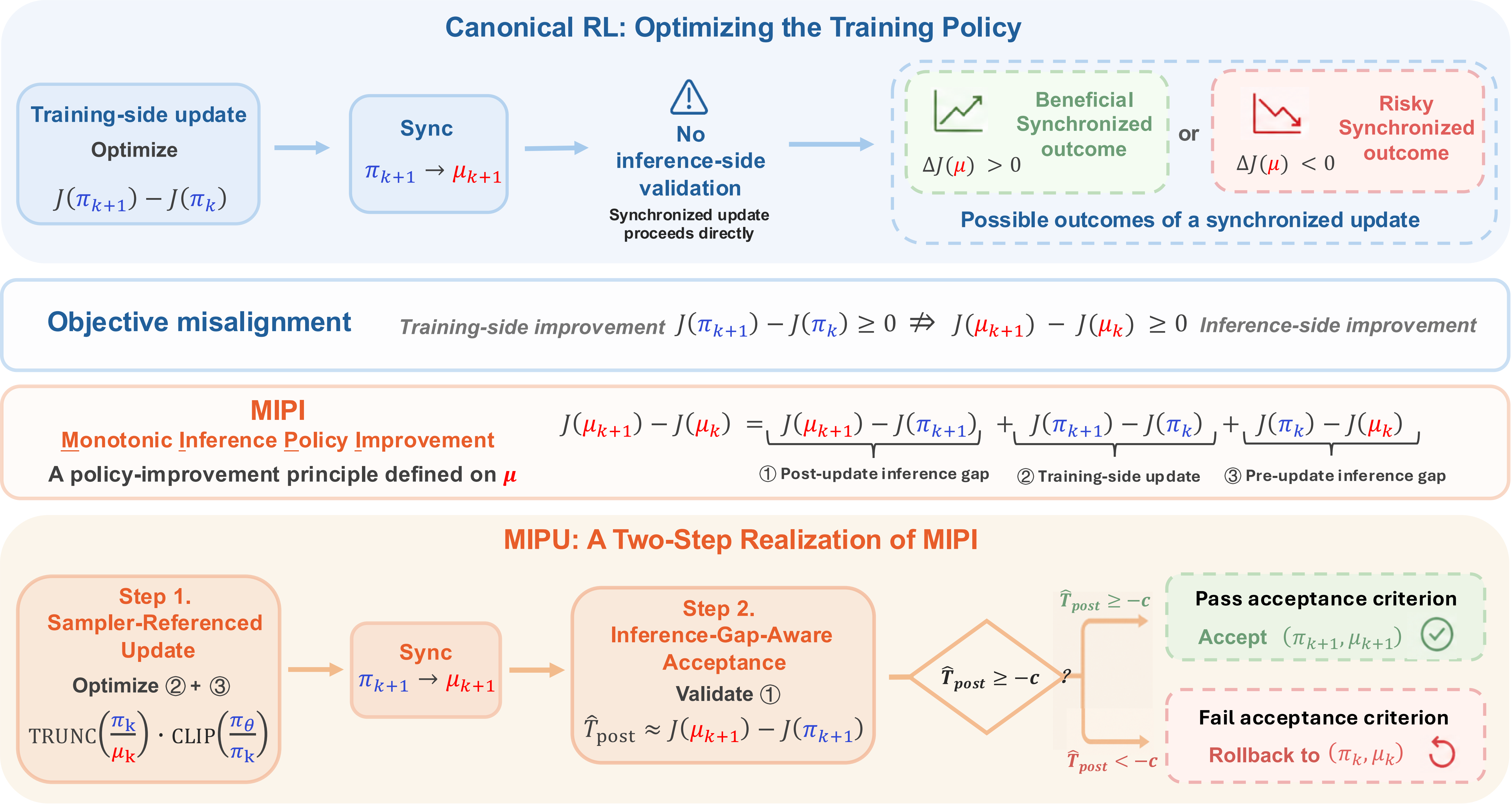
Canonical RL: Optimizing the Training Policy
Training-side update
Optimize
$J(\pi_{k+1}) - J(\pi_k)$
Sync
$\pi_{k+1} \rightarrow \mu_{k+1}$
No inference-side validation
Synchronized update proceeds directly
Beneficial Synchronized outcome
$\Delta J(\mu) > 0$
or
Risky Synchronized outcome
$\Delta J(\mu) < 0$
Possible outcomes of a synchronized update
Objective misalignment
Training-side improvement $J(\pi_{k+1}) - J(\pi_k) \geq 0 \not\Rightarrow J(\mu_{k+1}) - J(\mu_k) \geq 0$ Inference-side improvement
MIPI
Monotonic Inference Policy Improvement
A policy-improvement principle defined on μ
$J(\mu_{k+1}) - J(\mu_k) = \underbrace{J(\mu_{k+1}) - J(\pi_{k+1})}_{①} + \underbrace{J(\pi_{k+1}) - J(\pi_k)}_{②} + \underbrace{J(\pi_k) - J(\mu_k)}_{③}$
① Post-update inference gap
② Training-side update
③ Pre-update inference gap
MIPU: A Two-Step Realization of MIPI
Step 1. Sampler-Referenced Update
Optimize ② + ③
$\mathrm{TRUNC}\left(\frac{\pi_k}{\mu_k}\right) \cdot \mathrm{CLIP}\left(\frac{\pi_\theta}{\pi_k}\right)$
Sync
$\pi_{k+1} \rightarrow \mu_{k+1}$
Step 2. Inference-Gap-Aware Acceptance
Validate ①
$\hat{T}_{\mathrm{post}} \approx J(\mu_{k+1}) - J(\pi_{k+1})$
$\hat{T}_{post} \geq -c$?
$\hat{T}_{post} \geq -c$
Pass acceptance criterion
Accept (π_{k+1}, μ_{k+1})
$\hat{T}_{post} < -c$
Fail acceptance criterion
Rollback to (π_k, μ_k)